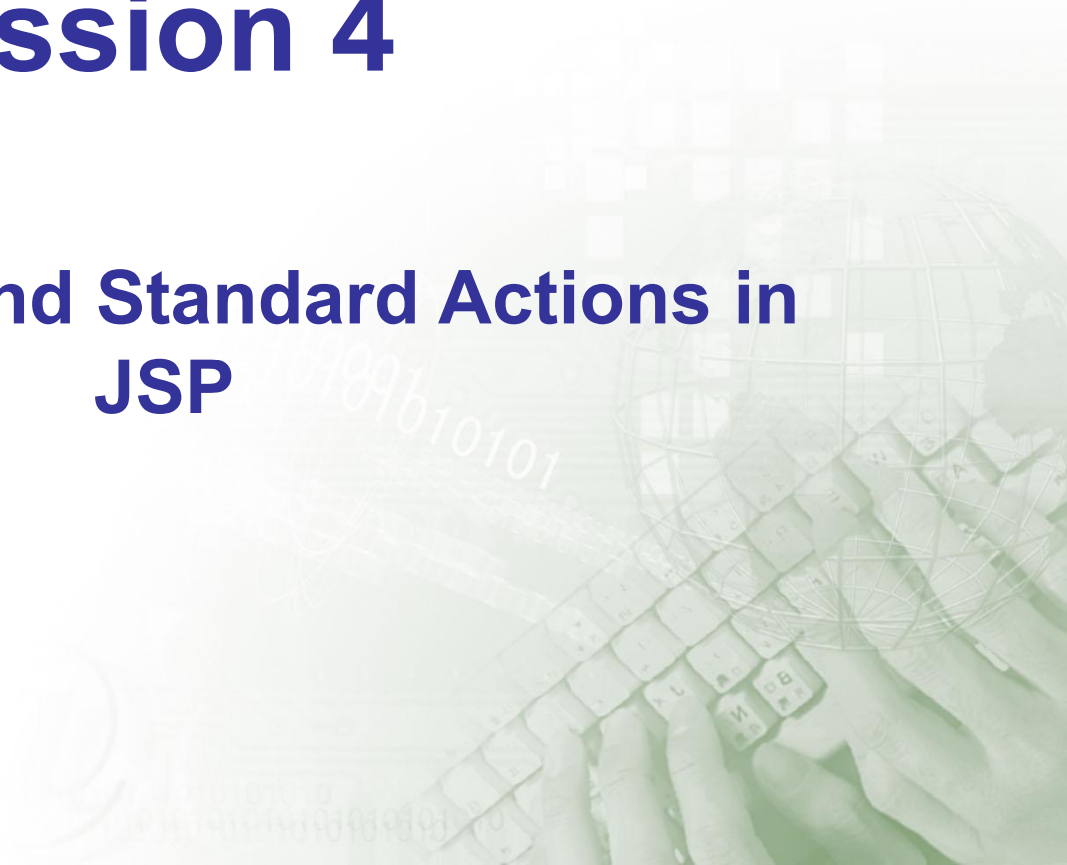


Session 4

JavaBeans and Standard Actions in JSP



Review 2-1

- ❑ JSP implicit objects are predefined classes that can be included in JSP expressions and Scriptlets.
- ❑ Implicit objects are included in a JSP page by JSP container.
- ❑ Input and output objects, such as `request`, `response`, and `out` objects are used to display and retrieve information from a Web page.

Review 2-2

- ❑ Scope communication and control objects, such as `session`, `application`, and `pageContext` provides a JSP page with access to the page context.
- ❑ The `page` object represents `Servlet` and `config` object that stores the initialization parameters of `Servlet`.

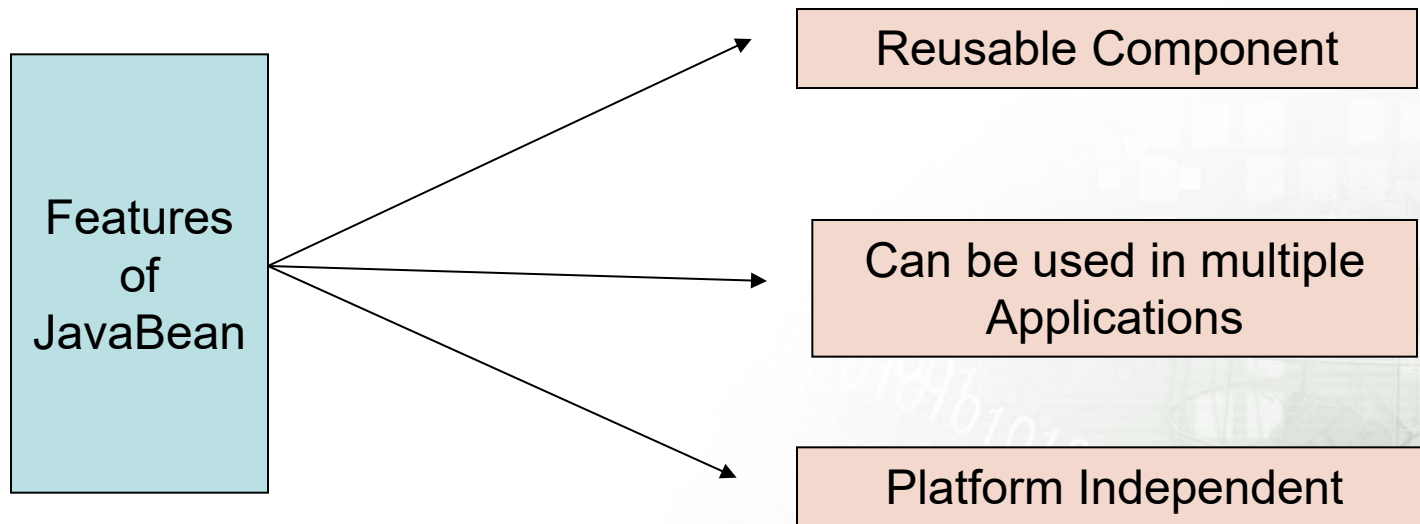
Objectives

- ❑ *Describe JavaBeans*
- ❑ *Describe various JSP Standard Actions*

Introduction to Java Beans

- ❑ A JavaBean is a specially constructed Java class written in the Java and coded according to the JavaBeans API specifications.
- ❑ It provides a default, no-argument constructor.
- ❑ It should be serializable and implement the **Serializable** interface.
- ❑ It may have a number of properties which can be read or written.
- ❑ It may have a number of "getter" and "setter" methods for the properties.

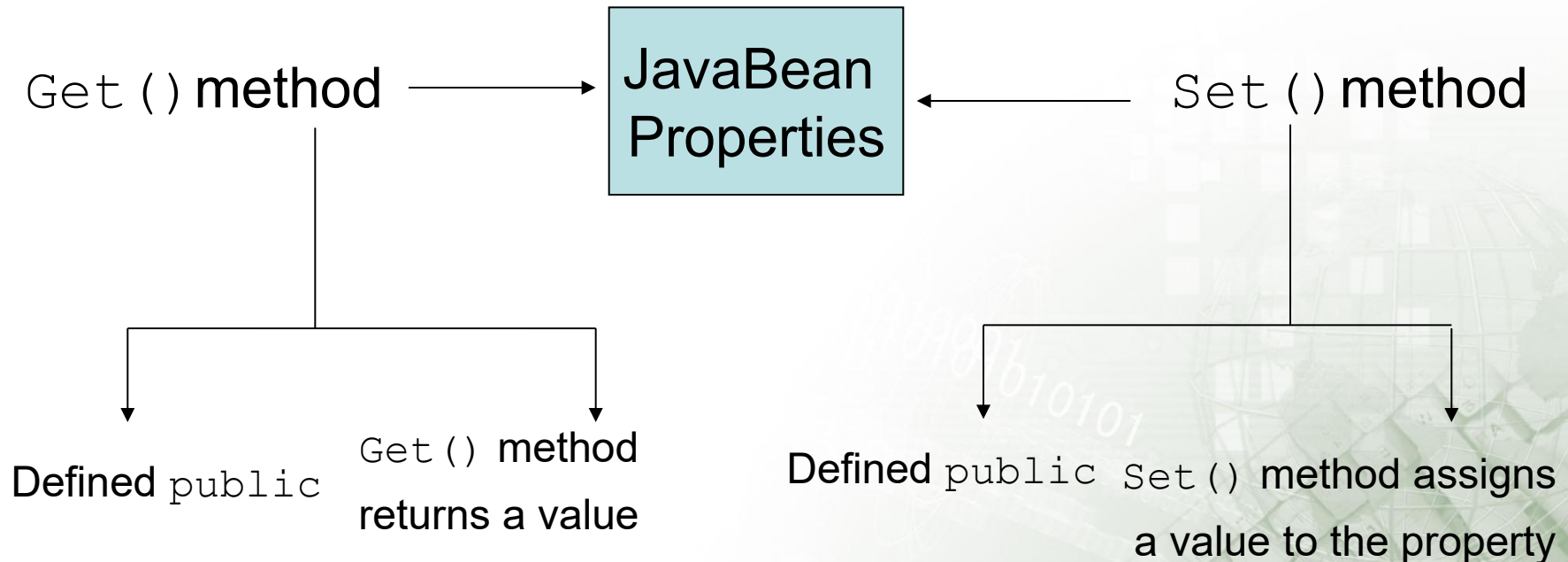
- ❑ JavaBeans are reusable components.



Get() and Set() Methods 2-1

- ❑ `get()` and `set()` methods are used to access properties of JavaBean.

To access JavaBean Properties



Get() and Set() Methods 2-2

```
import java.io.*;  
public class MyBean  
{
```

```
    private String name;
```

Property **name** is declared as private

```
    public MyBean()  
    {
```

Default constructor without arguments

```
    }
```

```
    public String getName()  
    {
```

```
        return name;  
    }
```

getter() method returns a value

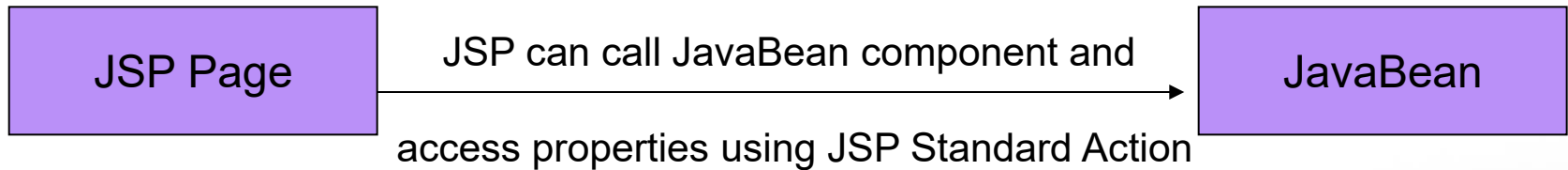
```
    public void setName(String myname)  
    {
```

```
        name = myname;  
    }
```

setter() method sets a value

```
}
```

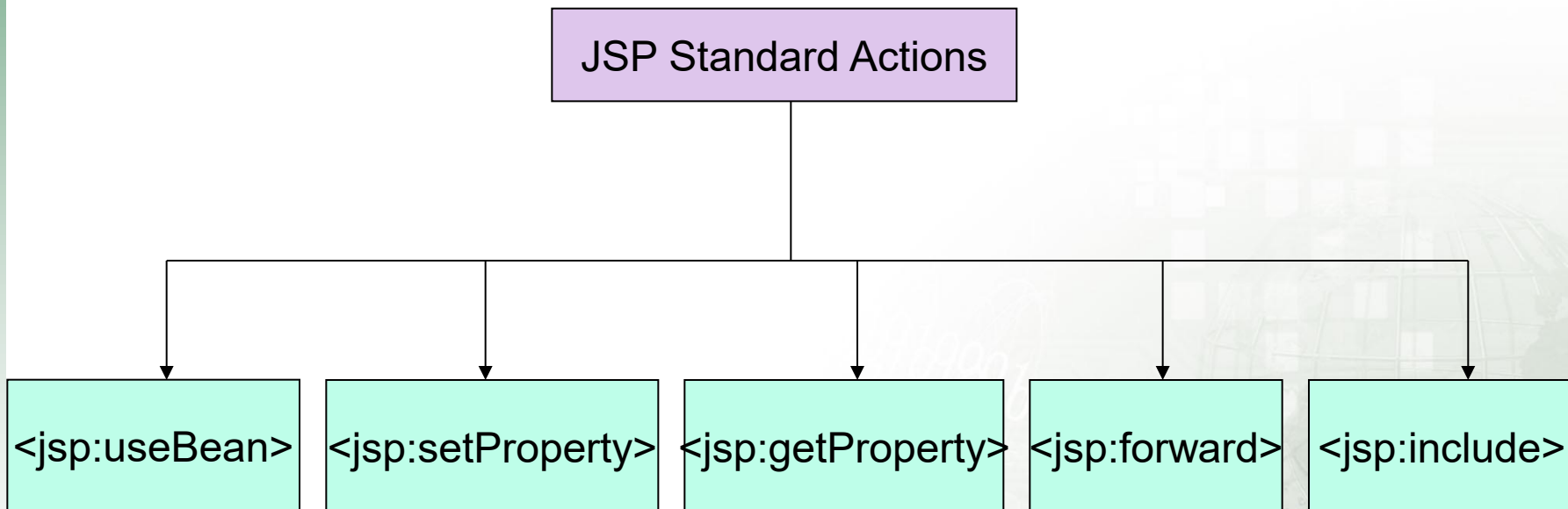

JSP Standard Action 2-1



- ❑ Standard Actions are used to:
 - ❑ Include JavaBean in a JSP page
 - ❑ Set and get properties of JavaBean
 - ❑ Forward the users request to another page
 - ❑ Include content of other page in current page
- ❑ Standard Actions can be an empty as well as a container tag.
- ❑ Attributes in Standard Actions are case-sensitive
- ❑ Standard Actions in JSP use `<jsp>` as prefix

JSP Standard Action 2-2

❑ The different JSP Standard Actions are:



useBean Action 2-1

...

<head>

```
<jsp:useBean id="BeanID" class="MyBean"  
scope="page"/>
```

</head>

...

Creates a reference and
includes an existing bean

Id attribute creates a
reference to the class
mentioned in `class`
attribute

useBean Action 2-2

- **Scope** specifies the availability of JavaBean in a JSP page. Different scope choices are:

Scope	Description
Page	The bean is available only till the page is in use. It gets destroyed when a new page is loaded.
Request	The bean exists till the user requests it.
Session	The bean remains in the session till it expires or gets deleted.
Application	The bean remains throughout the application.

setProperty Action

```
...  
<head>  
<jsp:useBean id="BeanID" class="MyBean"  
scope="session"/>  
<jsp:setProperty name="BeanID"  
property="name" value="Example"/>  
</head>
```

...

Name **attribute**
specifies reference to
a **JavaBean** class

Specifies property
name of a **JavaBean**

getProperty Action

```
...  
<head>  
    <jsp:useBean id="BeanID" class="MyBean"  
        scope="page"/>  
    <jsp:setProperty name="BeanID"  
        property="name" value="Example"/>  
</head>  
<body>  
    <jsp:getProperty name="BeanID"  
        property="name"/>  
</body>
```

Name **attribute** specifies
the reference to a
JavaBean class

Property **attribute**
specifies the property
name of a JavaBean

...

Creating JavaBean

❑ Usage of *get ()* and *set ()* methods in JavaBean

```
public class SimpleBean
{
    private String name = null;
    private int age = 0;
    public SimpleBean()
    {
    }
    public String getName()
    {
        return name;
    }
    public void setName(String username)
    {
        name = username;
    }
}
```

Using JavaBean in JSP

```
<html>
  <head>
    <title> Using JavaBean in JSP </title>
    <jsp:useBean id="BeanId" class="beans.SimpleBean"
scope="application"/>
    <jsp:setProperty name="BeanId" property="name"
value="Andrew"/>
    <jsp:setProperty name="BeanId" property="age"
value="25"/>
  </head>
  <body>
    Name is :<jsp:getProperty name="BeanId"
property="name"/> <br/>
    Age is :<jsp:getProperty name="BeanId"
property="age"/>
  </body>
</html>
```

Demonstration: Example 3

Forward Action 3-1

...

<body>

<jsp:forward page="Second.jsp"/>

</body>

...

Page attribute
specifies the
address of the other
page

Forward Action 3-2

- ❑ *Creating a JSP page*
- ❑ *Using `forward` action to forward the request of the user*

```
<html>
  <head>
    <title> Forward This Page </title>
  </head>
  <body>
    <jsp:forward
      page="RequestedPage.jsp"/>
  </body>
</html>
```

Demonstration: Example 4

Forward Action 3-3

```
<html>
  <head>
    <title>Request forwarded
    here</title>
  </head>
  <body>
    This page received a forwarded
    request from <br />
    <b>forward.jsp</b>,
    <br>This page is the output from
    <b>RequestedPage.jsp</b>,
    <br>but the URL is for
    <b>forward.jsp</b>
  </body>
</html>
```

Include Action 5-1

...

```
<body>
```

```
<jsp:include page="OtherPage.jsp"  
flush="true"/>
```

```
</body>
```

...

Page attribute
specifies the
address of the other
page

Include Action 5-2

❑ *Creating a JSP page that accept parameters and*

```
. . .  
<%  
    String tablename=request.getParameter("tablename");  
    String sqlquery=request.getParameter("sqlquery");  
    String whereclause=request.getParameter("ProductType");  
    out.println("<b>Sql query fired: </b>" + sqlquery + " " +  
tablename + " where ProductType='" + whereclause + "'<br/><br/>");  
    try {  
        Class.forName("sun.jdbc.odbc.JdbcOdbcDriver");  
        Connection con=DriverManager.getConnection  
("jdbc:odbc:pubs","sa","playware");  
        PreparedStatement s=con.prepareStatement(sqlquery+"  
"+tablename+" where ProductType='" + whereclause + "'");  
        ResultSet rs=s.executeQuery();  
        while(rs.next()) {  
            out.println(" " + rs.getString(1) + " " + rs.getString(2) + "  
$"+rs.getString(4)+"<br/>");  
        }  
    } catch (Exception e) {  
        out.println(e.toString());  
    }  
%>
```

Include Action 5-3

- ❑ *Creating a JSP page that includes a JSP page and sets parameters for the included page.*
- ❑ *Usage of `include` action and `param` element method*

```
<body>
    <jsp:include page="IncludeFileParameter.jsp"
flush="true">
        <jsp:param name="tablename"
value="ProductDetails"/>
        <jsp:param name="sqlquery" value="Select *
from"/>
        <jsp:param name="ProductType" value="Toys"/>
    </jsp:include>
    Above text is from IncludeFileParameter.jsp.
</body>
```

Demonstration: Example 7

Include Action 5-4

- ❑ *Creating an HTML page*
- ❑ *The page will be included in Example 9*

```
<html>
  <head>
    <title>Welcome to Car
    Shoppe</title>
  </head>
  <body>
    <h1 align="center">Welcome to Car
    Shoppe</h1>
  </body>
</html>
```

Demonstration: Example 8

Include Action 5-5

- ❑ *Creating a JSP page*
- ❑ *Using `include` action to include an HTML page*

```
<html>
  <head>
    <title>Using Includes</title>
  </head>
  <body>
    <jsp:include page="Welcome.html"
    flush="true"/>
    This text will appear after the
    contents of the <b>Welcome.html</b> are
    displayed
  </body>
</html>
```


Summary 2-1

- ❑ JavaBeans are reusable components that can be used in multiple applications.
- ❑ Bean properties can be accessed using the `get()` and `set()` methods.
- ❑ JSP pages can include JavaBean components using Standard Actions.
- ❑ JSP Standard Actions are used to insert files and bean components, and forward the user to another Web page.

Summary 2-2

- ❑ `useBean` action makes a `JavaBean` component available in a JSP page.
- ❑ `setProperty` action sets a value to the property of the bean.
- ❑ `getProperty` action is used to get a value from the bean.
- ❑ `Forward` action will forward the request of the user to another page.
- ❑ `Include` action is used to include a page in the current JSP page.